



The recall presentation.

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Question and purpose of analysis

- How has automotive reliability evolved over the past five years, and which manufacturers and components are driving the scale and frequency of these vehicle recalls?
 - What is the most apparent reason for recalls
 - What is the average amount of vehicles affected per recall
 - Which auto manufacturer is the least reliable (most # of recalls)
 - Has the amount of recalls gone up or down over the past 5 years
- The purpose of this project is to evaluate vehicle reliability and safety by identifying historical patterns in automotive recalls, looking at data from manufacturers and specific components.



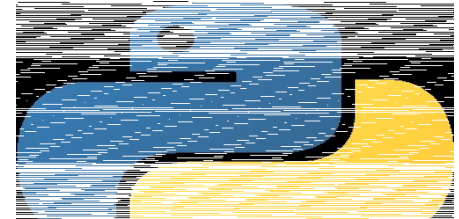
What datasets did I use?

- Recalls Data from [data.gov](https://catalog.data.gov/dataset/recalls-data)
 - Information in dataset includes recall info related to specific NHTSA campaigns, including data about reason for recall, manufacturer, amount of vehicles affected, and dates
 - Data was obtained via a CSV file
 - <https://catalog.data.gov/dataset/recalls-data>



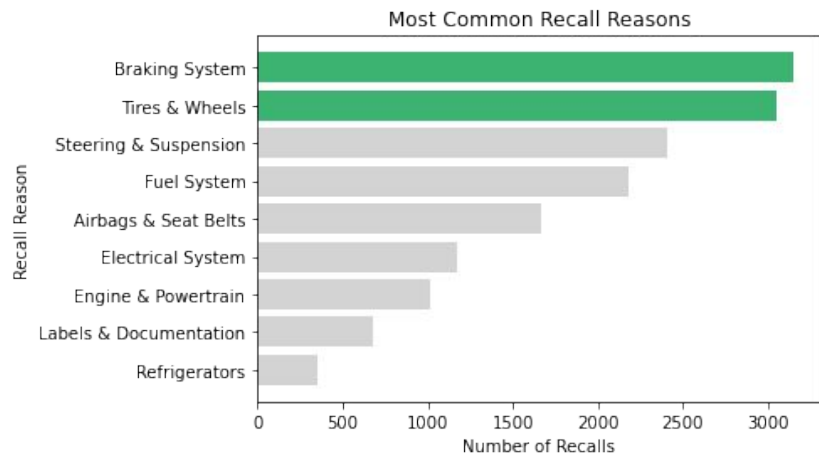
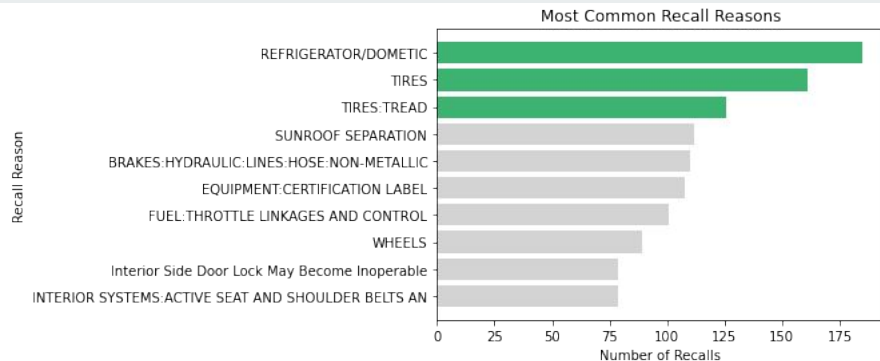
Methodology

- Clean dataset -> new CSV file
- An individual unit of analysis is recall case, indicated by an NHTSA ID
- Tools used:
 - Excel - View data
 - Jupyter Notebooks / Python - Execute code
 - Pandas - Data cleaning
 - Matplotlib - Visualizations



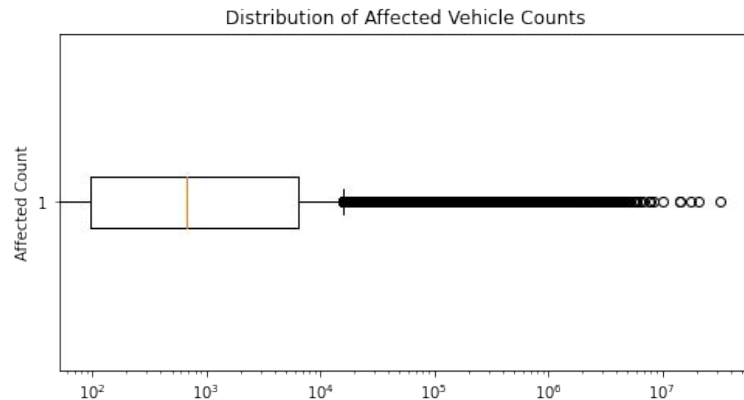
Results

- Question: What is the most apparent reason for recalls
- In the first visualization, RVs with fridges are the cause for the most recalls. However, since an RV is a very limited sample of the population, the most profound reason for recalls would be tires, more specifically tires that leak air.
- With the "umbrella" analysis, we can see that braking systems account for the most amount of recalls. Tires, however, are still prevalent and have slightly less issues than brakes, but still a lot relative to other categories. As for refrigerators, since RVs have such a low proportion of the population, they are last in recalls.



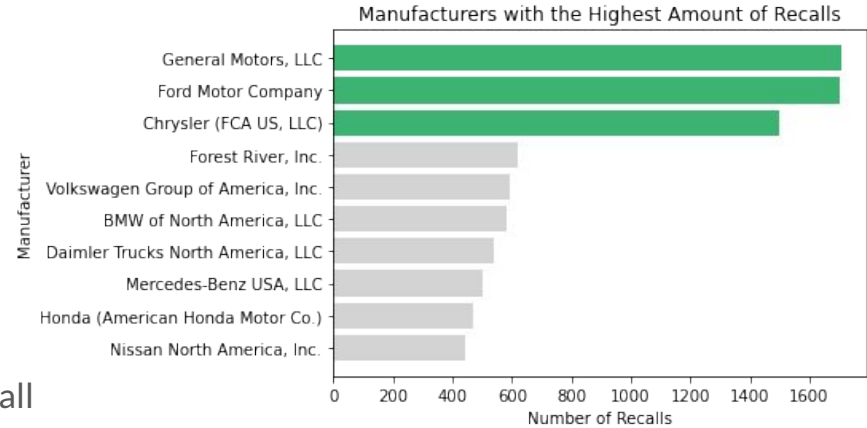
Results continued

- Question: What's the average amount of vehicles affected per recall?
- Average = 45295.46
- Median = 679
- It is shown that the upper fence of the dataset is around ~15k. This means anything above that number is an outlier, so we should rely on the median to more accurately depict how many vehicles are usually affected during recalls.



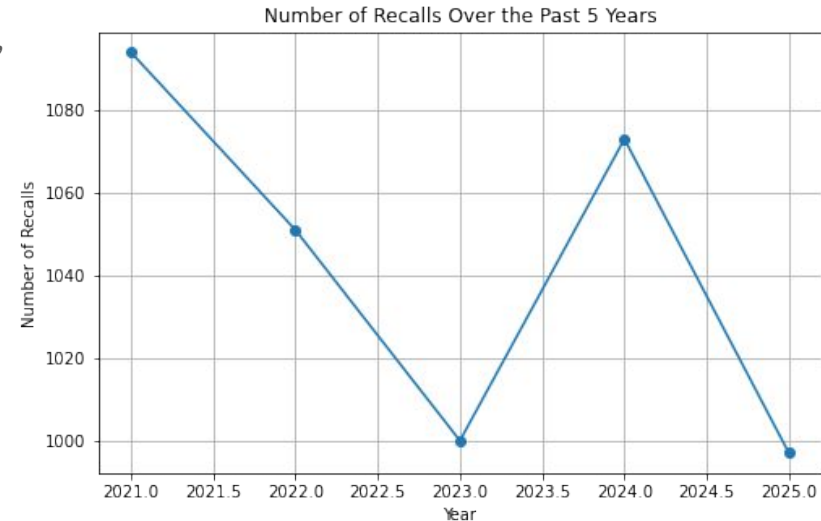
Results part 3

- Question: Which auto manufacturer is the least reliable?
- All the American companies have the highest recall rates. This, however doesn't necessarily mean they are the most unreliable. As all the other companies are based overseas (except Forest River, but they are an RV company), there is a high likelihood that recalls are more prevalent internationally, and they are able to fix the design before their vehicles come to the US market.



Results part 4

- Question: Has the amount of recalls gone up or down over the past 5 years?
- While there was a downwards trend from 2021-2023, and a spike in 2024, the amount of recalls generally is hovering around 1000-1100.





Future work

For the future, I would like to incorporate and use more datasets, including if these recalls had any direct involvement in accidents, as well as how the amount of recalls a company has put out affects their sales.



appendix/q&a

- Shoutout to Microsoft CoPilot
- <https://github.com/ccm-maxwellio/final-project/tree/main>

```
keep = ['report_date', 'nhtsa_id', 'manufacturer', 'subject', 'component', 'recall_type', 'affected_count', 'description', 'consequence', 'completion_rate']  
df = df[keep]  
df.columns
```